

* First-order logic :- (FOL)

- rep' in AI
- ① It is another way of knowledge
 - ② FOL is also known as predicate logic;
 - ③ It is a powerful language that develops info about the objects in a more easy way
 - ④ can also express the relationship b/w those objects;
 - ⑤ FOL does not only assume that the world contains facts like (PL) but also assumes objects

talks about only facts like T or F

here objects means = A, B, People, numbers, colors, houses - - ;

② Relationships :- sister of, brother of, has color, part of (unary relation)

③ functions :- father of, best friend of, end of ---, plus, one more than ---

(FOL) 2-parts :- ① syntax

② semantics

syntax :- Basic elements :-

predicate = part of a sentence containing a verb or stating something about the subject (attribute) i.e. logical statements that define properties of objects

① constants — 1, 2, 3, A, hyd, Sai - - - - - , numbers - - - - - (specific entities)

② variables — x, y, z, a, b - - - - - (place holders)

* ③ Predicates — Brother of, Father of, $>$, $<$, $\leq$, $\geq$ (describe relation)

④ functions — sqrt, ~~left leg of~~, location of (map object to object)

* ⑤ connectives — ~~(negation)~~ $\neg$, $\Rightarrow$ (Implication), $\wedge$ (and), $\vee$ (or), $\Leftrightarrow$ (double implies) say

⑥ equality — =

* ⑦ quantifiers — $\forall$ = for all (universal)

$\exists$ = there exists (existential)

$P \wedge Q$ = conjunction

$P \vee Q$ = disjunction

$P \Rightarrow Q$ = implication

$P \Leftrightarrow Q$ = equivalence

$\neg P$ = negation ✓

* In FOL, sentence can be rep' as ① atomic sentence ② complex sentence;

atomic sentence :- ① most basic sentence of FOL;

② these sentences are formed from a predicate symbol followed by parenthesis with a seq. of terms;

③ We rep' atomic sentences as $\Rightarrow$ predicate (term1, term2, term - - - n)

* First-order logic :- (FOL)

① It is another way of knowledge rep' in AI & it is an extension to (PL) {propositional logic}

② FOL is also known as predicate logic;

talks about only facts
like T or F

③ It is a powerful language that develops info about the objects in a more easy way

④ can also express the relationship b/t those objects;

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Eg:- T_1 Hari and T_2 Raghav are Brothers $\rightarrow$ predicate

Ans:- Brothers (Hari, Raghav) $\checkmark$

(2) Tommy is a dog $\rightarrow$ predicate

Ans dog (Tommy)

Types of predicate:-

(1) Unary - talks about single object

(2) Binary - defines a relationship two objects

Ex:- Brother (x, y) means x is the brother of y

(3) Complex sentences:- these are made by combining atomic sentences $\oplus$ connectives $\checkmark$
i.e. predicate ($T_1, T_2, T_3 \dots$) + connectives $\checkmark$

* FOL statements divided into 2 parts:- (either atomic sentence or complex sentence)

(1) subject - main part of statement;

(2) predicate - can be defined as a relation, which binds two atoms together in a statement;

Ex:- x is an integer;
first part (subject) second part (predicate)

so x = subject
is an integer = predicate $\checkmark$

* Quantifiers in FOL:-

(1) A quantifier is a language element which generates quantification;

(2) There are symbols that permit to det' (a) identify the range and ^{scope} of the variables in the logic expression;

(2) types (1) Universal Quantifiers (for all, everyone, everything) $\forall$ (range)

(2) existential (for some, at least one, someone) $\exists$ (scope)
(a, an, there is)

* Universal Quantifier :- (UQ)

It is a symbol of logical repⁿ, which specifies that the statement (within the range is true) for everything (8) every instance of particular thing;

* $\forall$ = symbol repⁿ $\Rightarrow$ the statement applies to (every element) in a given set;

* In UQ, we use implication symbol ' $\rightarrow$ '

say (Ex) :- If (x) is a variable, $\forall x$ is read as:

Ex :- $\forall x \in \mathbb{N}, x+x > x$
 say $x=1 \Rightarrow 1+1=2 > 1$
 $x=2 \Rightarrow 2+2=4 > 2$ } $\checkmark$

for all x $\checkmark$

for each x $\checkmark$

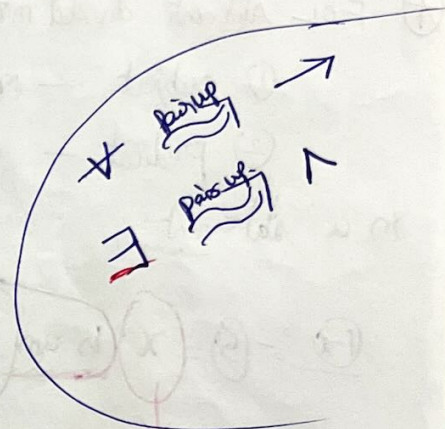
for every x $\checkmark$

Ex :- ① Everyman respects his parent

Ex :- $\forall x (Mammal(x) \rightarrow Respect(x, parent))$

② All mammals are animals;

$\forall x (mammals(x) \rightarrow animals(x)) \checkmark$



* Existential Quantifier (E) :-

It is a type of quantifier, which express that the statement (within) its scope is true for atleast one instance of something;

* $\exists$ = symbol repⁿ $\Rightarrow$ the statement (is) true for (atleast) one element in a given set;

* {here, we always use AND (8) Conjunction symbol ($\wedge$)}

say (Ex) If x is a variable, then $\exists(x)$ is read as:

Ex :- ① some boys play cricket

Ex :- $\exists x (Boy(x) \wedge playcricket(x))$

② A student passed

Ex :- $\exists x (student(x) \wedge passed(x))$

there exist a(x) $\checkmark$

for some (x) $\checkmark$

for atleast (x) $\checkmark$

Ex :- $\exists x \in \mathbb{N}; x^2 = 25$
 say $x=1 \Rightarrow 1^2 \neq 25$ but $x=5 \Rightarrow 5^2 = 25$ $\checkmark$
 $x=3 \Rightarrow 9 \neq 25$

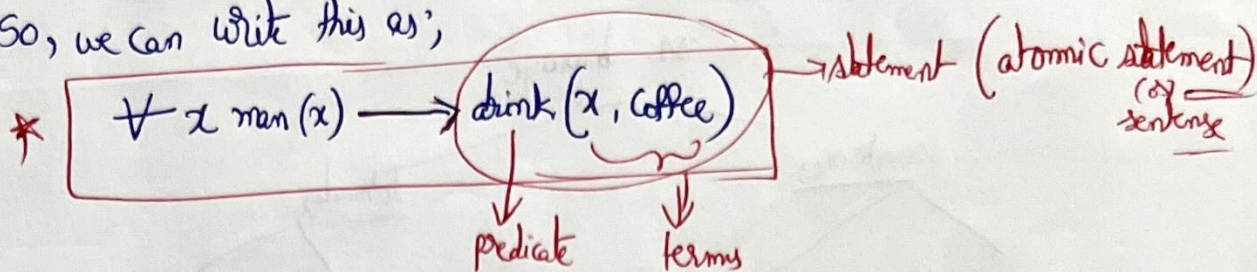
atleast so;

Ex: "All man drink coffee"

Q: let x is a variable

$\therefore x_1 \text{ drink coffee} \wedge x_2 \text{ drink coffee} \wedge x_n \text{ drink coffee} \dots$

So, we can write this as;



Finally; there are all x where $x = \text{man}$, who drink coffee ✓

Ex ② All birds fly

Q: $\forall x \text{ bird}(x) \rightarrow \text{fly}(x)$
 $x = \text{bird}$
 $\rightarrow$ predicate

③ Every man respects his parent

Q: here, predicate is "respect(x, y)"
 $x = \text{man}$
 $y = \text{parent}$

$\forall x \text{ man}(x) \rightarrow \text{respects}(x, \text{parent})$ (or) respects(x, y) ✓

④ Some boys are intelligent;

Q: $x_1 \text{ intelligent} \vee x_2 \text{ intelligent} \vee x_3 \text{ intelligent} \dots$

so;

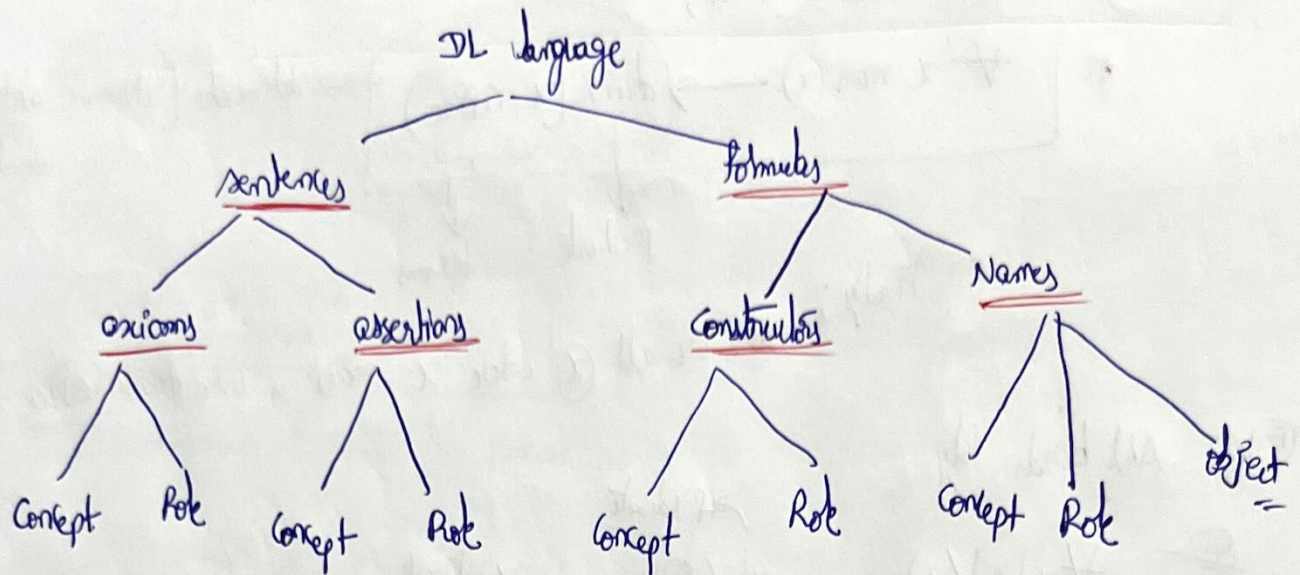
$\exists x \text{ boy}(x) \wedge \text{intelligent}(x)$ $\downarrow$ means

there are some boys x where x is a boy who is intelligent ✓

* Description Logic :- ① also known as logic of noun phrases;

② FOL is a semi-decidable but DL is a decidable subset of FOL;

③ DL designed for knowledge representation @ ontologies;



④ more specific constraints make it possible to model more specific forms of inference;

⑤ DL Represent knowledge about :-

- ① { Categories
- { Individuals
- { Relationships

② Terminology

③ TBox = the portion of knowledge base contains the terminology

④ ABox = " " ; facts about individuals

⑤ ontology = Hierarchical Representation of subset/superset relations among categories ✓

⑥ DL is a family of knowledge representation languages which can be used to rep^l the terminological knowledge of an application domain in a structured @ formally well understood way.

* Core components of DL :-

- (1) Concepts (C) — classes (2) sets of individuals (eg., person, animal, car).
- (2) Roles (R) — binary relations b/t individuals (eg., owns, has child).
- (3) Individuals (a, b, c) — specific objects (eg., John, Sai, Car).

* Constructors of DL :-

Constructors	Symbol	Example
atomic Concept	A	human
atomic Role	R	has-child
Conjunction	$C \sqcap D$	Human $\sqcap$ male
Disjunction	$C \sqcup D$	doctor $\sqcup$ lawyer
negation	$\neg C$	$\neg$ male
subsumption	$\subseteq$	person $\subseteq$ animal { means every person is an animal }

Ex: On the set of all those parents having a male child who is a doctor or a lawyer is exp' as

Ans: Has-child.male $\sqcap$ (doctor $\sqcup$ lawyer)

② Sai is a person who has atleast one pet that is a dog;

Ans: Sai: person $\sqcap \exists$ haspet.Dog ✓

③ "A mother is a woman who has a child"

Representing (Name) phrases

Ans: mother $\equiv$ woman $\sqcap \exists$ haschild.person ✓

④ Every student reads a book

} $\Rightarrow$ sentence semantics ✓

Ans: student, $\subseteq \exists$ reads.book

uses of words in an action ✓

① Syntax-Driven Semantic Analysis :- (semantic) means finding meaning of sentence;

② It is a stage in (NLP) that links syntactic struc. (grammar) with semantic (meaning)

③ It uses syntactic parse tree as the backbone for constructing (semantic) representation ensuring that the meaning of a sentence is derived compositionally (fully);

④ It is based on 'principle of compositionality',

∴ the meaning of sentence is a funcⁿ of the meanings of its parts & the way they are syntactically combined; so, the syntax tree structure guides, how semantic rep^l are combined - word by word (bottom up)

⑤ It uses, syntax analysis to guide the process of arranging semantic representⁿ ation

⑥ what is syntax-driven semantic analysis :-

- the process of generating semantic representations (meaning struc.) directly from syntactic struc.

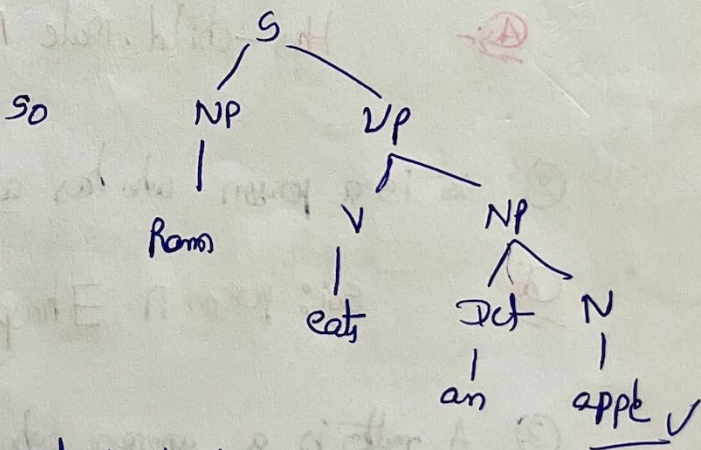
it involves :-

- (1) parsing
- (2) semantic rule attachment
- (3) semantic composition

(Ex) :- S = "Ram eats an apple"

① parsing :-

$S \rightarrow NP + VP$
 $VP \rightarrow V + NP$
 $NP \rightarrow Det + N$
 (a)
 pronoun



② semantic rule attachment :-

Grammar Rule

$S \rightarrow NP + VP$

$VP \rightarrow V + NP$

each grammar Rule is attached with semantic rule ✓

Semantic Rule

$S.sem = VP.sem(NP.sem)$

$VP.sem = V.sem(NP.sem)$

NP $\rightarrow$ Noun phrase

NP $\rightarrow$ Det + N

$v \rightarrow \text{eat}$

$$NP \cdot \text{Sermon} = \text{Ramm}$$

NP. term = apple

$$\boxed{v.x_{\text{rm}} = \lambda x.\text{eat}(\text{Ram}, x)}^*$$

λ = lambda calculus is used to represent functions & application during meaning composition

where $\lambda x =$ Represents a func' (lambda exp')

eat (Ramm, x) means Ramm ~~eat~~ eats something

③ semantic composition :-

Now meanings are combined according to the parse tree; to form
 $\Rightarrow$ Ramon ; meaning = Ramon logical Repent.

Step 1:- $NP \rightarrow \text{Ramen}$; meaning = Ramen

"(2) :- NP $\rightarrow$ an apple ; meaning = apple

11 (3) :- $V \rightarrow \text{eat}$; meaning = $\lambda x. \text{eat}(\text{Ramu}, x)$

"(4)":- apply the verb meaning to the object; eat (Ram apple) ✓

↓
this rep' logical meaning of
sentence ✓

Analys :- syntactic element

semantic Role

NP $\longrightarrow$ Refers to entities (individuals, objects)

VP → Refers to actions, events & relation

PP $\rightarrow$ adds relations & properties

S (sentence) $\rightarrow$ asserts a proposition about the world

* why syntax-driven? - The syntactic structure determines who does what to

i.e. how sentence grammar (syntax) tells us the roles of words in an action ✓

⊛ Number - Derived romantic Ambiguity ⊛

⊛ Let's say :-

① dog bites man
② man bites dog;

} words are seemed to be same but semantic are different

<u>sentence</u>	<u>who does action</u>	<u>who receives action</u>
①	dog	man
②	man	dog

⊛ word order det' (who does what to whom);

Word senses :- simply → It is a specific meaning of a word det' by the context in which it is used;

Word sense = the right (or) correct meaning of a word in a given sentence ✓

— many english words have more than (one meaning); the right meaning dep' on the context (or) surrounding words;

Ex:- Bow, lead, Row, light, bark, nail, novel, season, bank, bat

Bow :- (n) — she ties her hair up with a ribbon ⊕ made a little (Bow);

(v) :- when mahima meets her grandparents, she always (bows) to greet them

Types of word senses :-

① polysemy :- single word has multiple related meanings; the meanings are connected ⊕ derived from the same concept;

Ex:- ① Head of a person

Head of a table → top position ✓

" of the dept;

② Homonymy :- this occurs when ② words have the same spelling ⊕ pronunciation but completely different meanings;

Ex:- bark :- (n) - tree bark is used to make paper;

(v) - I have a dog myself, but his dog barked so viciously that I ran away, frightened;

(ii) season :- (n) - my fav. season is winter, because the snow is so magical

(v) - the recipe says to season the dish with plenty of salt;

(iii) Novel :- (N) - the grapes of Wrath is veravi's best novel;

(adj) - the steam engine was a novel invention that changed the way people travelled;

(3) Homophones :- these are the words that sound the same but have different meanings (or) spellings

Ex:- (1) Flower -

(2) Flour - powder using in cooking ✓

(4) Homographs :- these are the words that have same spelling but diff' meanings (sometimes different pronunciation)

Ex:- Lead :- to guide

Lead :- a metal element ✓

(5) Synonyms :- different words have the same or very similar meanings

Ex:- big - large

Happy - Joyful

Begin - start;

(6) Antonyms :- two words have opposite meanings;

Ex:- Hot - cold

Big - small

Up - down ✓

* Semantic Relations :-

* Relation between senses :- (lexical-semantic relations)

- there are links b/w sense nodes in resources like wordnet;
- useful for similarity & disambiguation;

Ex :-

① synonymy :- car $\cong$ automobile

② Antonymy :- hot $\leftrightarrow$ cold

③ Hypernymy (is a) :- dog $\subseteq$ Canine $\subseteq$ animal ✓

④ meronymy (part of) :- wheel ; part of car;

⑤ Entailment (verbs) :- snore entails sleep;

⑥ derivationally related forms :- decide $\leftrightarrow$ decision ✓

Semantic Roles & Relational Restrictions :-



(E) (also called semantic Roles) → tells about the role each participant plays in the action

(C) described by a (verb) in a sentence;

In simple words, they answers questions like :-

who did the action?

|| received the action?

How was the action done?

where / when did it happen?

②

① Agent → the person (or) thing that performs the action;

Ex :- harsha playing the cricket;
Agent

② Patient/Theme :- the entity that is affected by the action;

(Ex) :- vasavi ate the (apple) → patient ✓

③ Experiencer :- the person who experiences a feeling, perception or emotion;

(Ex) :- (vasavi) fears spiders

④ Instrument :- tool used to perform the action;

(Ex) :- vasavi cut the cake with a (knife);

⑤ Beneficiary :- the person who benefits from the action;

(Ex) :- Janu baked a cake for (Ram) ✓

↑ also Recipient;

⑥ Goal - destination where something moves/ends up;

(Ex) :- she walked to (park) ✓

⑦ Source - starting point _{from} which something moves

(Ex) :- The train left the (station)

⑧ location & time also;

Pragmatically :- TR are semantic roles that describe how different participants in a sentence are related to the action expressed by the (verb);

① "John (cut) the apple with a knife in the kitchen yesterday"

↓ ↓ ↓ ↓ ↓

agent patient instrument location time

② Vasavi gave barsha a book at school

Agent Give (verb) Recipient Theme Location

∴ (verb) is the center ③ Thematic roles describes how the utterance is related to that (verb) ✓

* Selectional Restrictions :- (SR's)
 In NP, verbs often expect their argument (s, o) to belong to certain semantic categories. To this 2-concepts talks
 (1) selectional restrictions
 (2) " preferences;

① selectional restrictions :-

These are strict semantic constraints that a word (verb) places on the types of words that can appear as its arguments;

Ex:- Vasavi eat an apple; ✓
 ↓
 Verb:-

<u>Role</u>	<u>expected type</u>
Agent	animate (human)
object	food

The store ate an apple X
 ↓
 not animal so it violates the (S.R)'s;

② SP's :- These are soft or probabilistic constraints where a word prefers certain semantic categories but can sometimes accept others;

* Ex:- The car drank petrol
 → actually drink prefers animate ~~sub~~ subjects

So preferred ex:-

Vasavi drank water
 The child drank milk

But we sometimes say :- The car drank petrol;

WSD:-
the computational process of finding the correct meaning (sense) of a
polysemous word based on its context;

Imp. for:- ① MT's ② Info Retrieval ③ Chatbots ④ Virtual assistants
④ sentiment ⑤ context analysis

Approaches:- ① knowledge based — uses lexical resources like Wordnet
② supervised learning — trains on labelled sense;
③ unsupervised learning — cluster words based on context similarity;

Lesk algorithm steps ① identify the ambiguous word
② Retrieve its possible senses from Wordnet;
③ Compare each sense's gloss with context words
④ The sense with the highest overlap is
Chosen;

presenters, CSE-II 7th period

K₆, K₇, K₂, L₁, L₂, L₀, L₅, L₆, M₀, M₁, M₂, M₅, M₆, M₉, N₁, N₂, N₄, N₆, N₇,

O₀, O₁, O₆, O₅, O₈, O₉, P₀, P₃, P₄, P₅, P₆, P₇, P₉, Q₀, Q₃, Q₄, Q₅,

Le' - Nil

WordNet :- A lexical database for English ; used to perform NLP tasks ;

- ↓
- 10k nouns
 - 11k verbs
 - 22k adj.
 - 4.5k adverbs
- * developed @ Princeton university @ 1986 ;
 - * organizes words into synsets (sets of cognitive synonymy)
 - * Rep^t semantic relationship b/t words ; such as

- Hyponymy
- synonymy
- Antonymy

- * Structure of WordNet :-
- ① words are grouped into synsets ;
 - ② synsets are connected via semantic relationship ;

Ex :- synsets :- { dog, domestic dog, canis familiaris }

Hyponym :- { Canine , Carnivore }

- * WordNet components :-
- ① Nouns, verbs, adj, adverbs . . .

- ② synsets :- Group of synonymous words

- ③ semantic relations :-

Hyponymy
synonymy @ Antonymy

- * Importance :-
- ① provides structured semantic knowledge for english ;

- ② widely used in NLP ; like ① WSD, ② Text Classification, ③ Clustering

- ③ QA & S

- ④ IR

- ③ acts as a thesaurus @ ontology simultaneously ;

* Word Similarity ^(1/5) - it measures how close in meaning 2 words are;

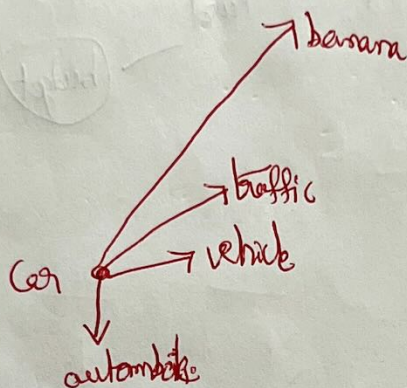
- measuring WS :- (1) Thesaurus based similarity $\rightarrow$ using wordnet ^{gives similarity score directly}
(2) distributional similarity $\rightarrow$ vector space similarity

* "semantic distance" is an indication of word similarity;

Ex: $\text{synonyms} < \text{similar words} < \text{related words} < \text{unrelated words}$

$\boxed{\text{car}} \Rightarrow \text{automobile} < \text{vehicle} < \text{traffic} < \text{banana}$

* This distance can be used to create word classes (clusters); useful for smoothing



We can measure dist (or) similarity;

- distance $\text{dist}(w, w')$ is big $\Rightarrow w, w'$ are unrelated;

- similarity $\text{sim}(w, w')$ is big $\Rightarrow w, w'$ are similar;

if $0 \leq \text{dist} \leq 1 \Rightarrow$

$$\text{define } \text{sim}(w, w') = 1 - \text{dist}(w, w')$$

$\text{car} \leftrightarrow \text{automobile} = \text{highly similar}$

$\text{car} \leftrightarrow \text{road} = \text{moderate}$

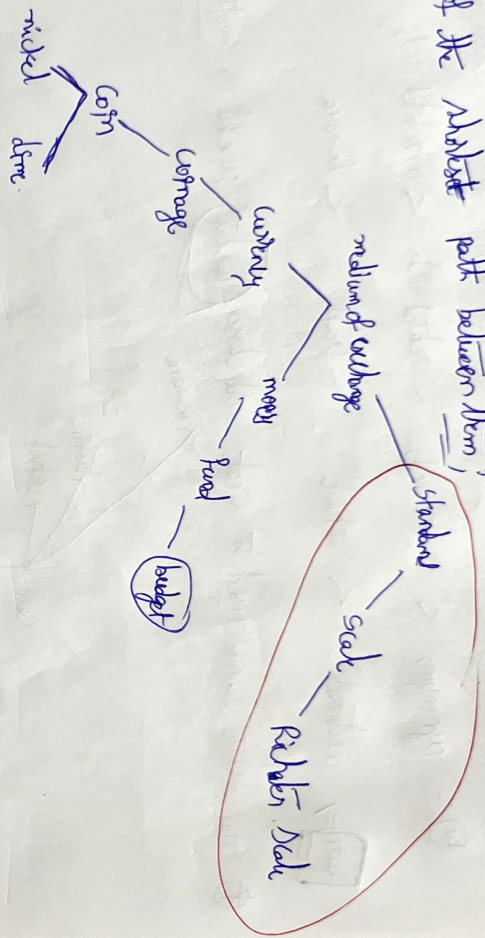
$\text{car} \leftrightarrow \text{banana} = \text{not similar}$

$$\begin{aligned} & \text{(or)} \\ & = \frac{1}{\text{dist}(w, w')} \end{aligned}$$

① the taxonomy-based MS

the taxonomy is like a webster, where the words are organized into semantic relations, like synonyms, hyperonyms ...

② the path length (dist) $pathlen(5, 5')$ between ② - server 5, 5' is the length of the shortest path between them;

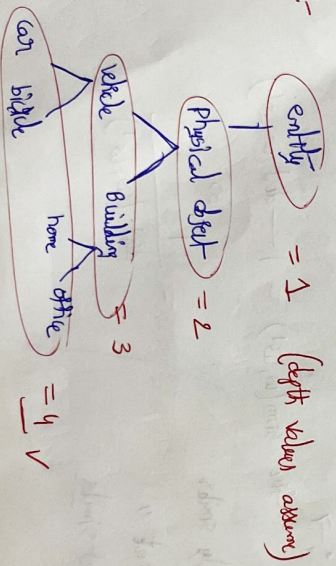


for above :- pathlen (nickel, dime) = 2

pathlen (nickel, money) = 5

pathlen (nickel, budget) = 7

③ :-



entity = 1 (depth values assume)

physical object = 2

vehicle = 3

office = 4 ✓

* shortest path b/w car & bicycle :- (2) edges i.e

Car $\rightarrow$ vehicle $\rightarrow$ bicycle

① path based similarity :-

$$\text{similarity} = \frac{1}{\text{shortest path length}}$$

$$\text{sim}(\text{car}, \text{bicycle}) = \frac{1}{2} = 0.5 \quad \text{i.e. shorter path} \Rightarrow \text{higher similarity}$$

② Wu-Palmer similarity :-

$$\text{sim} = \frac{2 \times \text{depth}(\text{LCS})}{\text{depth}(w_1) + \text{depth}(w_2)}$$

{ It considers both depth @ hierarchy }

LCS = least common subsumer

= (common ancestor in hierarchy)

$\text{LCS}(\text{car} \& \text{bicycle}) = \text{vehicle} \Rightarrow \text{depth} = 3$

$$\text{SO } \text{sim}(w_1, w_2) = \frac{2 \times 3}{4 + 4}$$

$$= \frac{6}{8} = 0.75$$

③ Leacock-Chodorow similarity (LCH) :-

$$\text{sim} = -\log_{10} \left(\frac{\text{path length}}{2 \times \text{max depth of hierarchy}} \right)$$

$$\left. \begin{array}{l} \text{path length} = 2 \\ \text{max depth} = 4 \end{array} \right\} \text{sim} = -\log_{10} \left(\frac{2}{8} \right) = 0.602 \checkmark$$

out of above 3 $\Rightarrow$ case ② gives highest value;

* path based easy but less accurate $\checkmark$

* W.S. using distributional methods :-

① hypothesis :- words that occur in similar context have a similar meaning

② co-occurrence matrix :-

consider a small corpus :-
 " sark Engg. college bhimavaram "
 " vasavi " " Tadepalligudem "

③ Build vocabulary :-

context words = { Engg, college, bhimavaram, gudem }

④ Create vectors :-

<u>word</u>	<u>engg</u>	<u>college</u>	<u>bhim.</u>	<u>gudem</u>
sark	1	1	1	0
vasavi	1	1	0	1

⑤ Cosine similarity :-

$$\text{similarity} = \frac{A \cdot B}{|A| |B|}$$

~~dog~~ (1, 0)

$$\text{sark} = (1, 1, 1, 0) = A \Rightarrow |A| = \sqrt{3}$$

$$\text{vasavi} = (1, 1, 0, 1) = B \Rightarrow |B| = \sqrt{3}$$

$$A \cdot B = (1 \times 1) + (1 \times 1) + 0 + 0 = 2$$

$$\text{so similarity} = \frac{2}{\sqrt{3} \cdot \sqrt{3}} = \frac{2}{3} = 0.667 \checkmark$$

∴ close to 1 = high similarity
 close to 0 = low "

⊛ Brill's Tagger :- It is a POS tagging algorithm that assigns grammatical categories to words in a sentence;

- It uses a method called Transformation-based learning (TBL);

- Its hybrid approach :- ① starts with simple tagging

② Improve using learned linguistic rules

- It mimic how humans correct mistakes step-by-step;

Core Idea

TBL :- starts with rough guesses $\rightarrow$ learn rules $\rightarrow$ fix errors iteratively

Step ① :- initial state tagging :- using most frequent tag (v) a simple lexicon

② Compare with correct tags;

③ generate candidate rules $\Rightarrow$ (like) change $NN \rightarrow VB$ if prev. word is NN
" $NN \rightarrow VB$ if word ends with S
 $NN \rightarrow VB$ " " " " ing

④ score rules $\Rightarrow$ each rule is evaluated :-

$$\text{score} = \text{errors corrected} - \text{errors introduced};$$

⑤ select best rule :- choose rule with highest score & apply it;

⑥ Repeat iteratively :-

Pro's :-

① easy to understand

② Req. less data than DL

③ linguistically meaningful

④ Good accuracy

Ex:- "I want to book a flight"

Step 1:-
noun verb | noun and NN
↓ ↓ | ↓ | ↓
PRP VB TO NN DT NN

②:- PRP VB TO VB DT NN

③:- change NN → VB if previous tag is TO

NN → VB if word is "book"

NN → VB if previous word is (to)

④:- each rule is tested on training data

Rule 1:- NN → VB if prev. tag is TO

fixes:- "to book" introducing errors? very few

Score = high

Rule 2:- NN → VB if word = "book" ; works only for one word so

Score = low

Rule 3:- similar to Rule 1 but less general so Score = medium

⑤:- select Best score i.e. Rule 1

⑥:- apply Rule

⑦ Repeat (if needed)